

Product Backlog

Software Engineering - SVNC.00.308

Project: StayEase Hotel Reservation System
Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustuhov)
Date: March 2, 2026
Version: 1.0

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User Stories

ID	User Story	Acceptance Criteria	Priority	Est.
US-01	As a hotel guest, I want to search for available rooms by dates and guest count, so that I can find a room that fits my travel plans	AC1: Guest can enter check-in date, check-out date, and number of guests; AC2: Only rooms available for the full date range are shown; AC3: Results display room type, price per night, and capacity	Must-have	M
US-02	As a hotel guest, I want to view detailed information about a room, so that I can decide if it meets my needs	AC1: Room page shows photos, description, amenities, and nightly rate; AC2: Current availability status is displayed for selected dates	Must-have	S
US-03	As a hotel guest, I want to register and log in with my email and password, so that I can make and manage reservations	AC1: Guest can register with email, password, full name, and phone; AC2: Guest can log in with email and password; AC3: Invalid credentials show clear error message	Must-have	M
US-04	As a logged-in guest, I want to book an available room for my selected dates, so that I can secure my stay at the hotel	AC1: Booking captures room, dates, guest count, and special requests; AC2: Room availability updates in real-time after booking; AC3: Confirmation page displays reservation ID and summary	Must-have	M
US-05	As a system, I want to prevent double bookings for the same room on overlapping dates, so that no two guests are assigned the same room	AC1: Booking attempt for an occupied room fails with clear message; AC2: System suggests alternative rooms of the same type; AC3: Concurrent booking attempts are handled safely	Must-have	S
US-06	As a hotel guest, I want to cancel my reservation at least 24 hours before check-in, so that I can free the room if my plans change	AC1: Cancel button is available for reservations >24h before check-in; AC2: Cancel button is disabled within 24 hours of check-in; AC3: Cancelled room returns to available inventory immediately	Must-have	M
US-07	As a hotel guest, I want to view my past and upcoming reservations, so that I can keep track of my booking history	AC1: All reservations shown with room type, dates, price, and status; AC2: Reservations sorted by date (upcoming first); AC3: Status badges displayed (Confirmed, Cancelled, Completed)	Must-have	S

US-08	As a hotel guest, I want to modify my reservation dates or room type, so that I can adjust my plans without cancelling	AC1: Guest can change check-in/check-out dates subject to availability; AC2: Guest can switch room type if alternative is available; AC3: Updated total price is shown before confirmation	Must-have	L
US-09	As front desk staff, I want a dashboard showing today's check-ins, check-outs, and occupancy, so that I can manage daily operations efficiently	AC1: Dashboard displays today's arrivals and departures count; AC2: Current occupancy rate is shown as percentage; AC3: Searchable list of all active reservations is available	Must-have	M
US-10	As front desk staff, I want to mark guests as checked-in or checked-out, so that room status reflects actual occupancy	AC1: Staff can click Check-In for arriving guests; AC2: Staff can click Check-Out for departing guests; AC3: Room status updates accordingly (occupied/vacant)	Nice-to-have	M
US-11	As a hotel guest, I want to receive email notifications for booking events, so that I have written confirmation of my reservation	AC1: Confirmation email sent on booking creation; AC2: Notification email sent on modification or cancellation; AC3: Reminder email sent 24 hours before check-in	Nice-to-have	L
US-12	As the hotel manager, I want to view occupancy reports by date range, so that I can analyze booking trends and plan staffing	AC1: Report shows occupancy rate by room type; AC2: Report covers a selectable date range; AC3: Average length of stay and total reservations displayed	Nice-to-have	M

Estimation Key: S (Small, 3 pts, 15-25 min) | M (Medium, 5 pts, 30-45 min) | L (Large, 8 pts, 50+ min)

Prioritization Rationale

Business Value

US-01 through US-04 (Search, View, Login, Book) are highest priority because they form the core booking workflow. Without these, the system has no functionality. A guest must be able to search rooms, view details, authenticate, and make a reservation for the product to deliver any value.

Technical Dependencies

US-03 (Registration/Login) must be completed before US-04 (Book Room) because booking requires an authenticated user. US-01 (Search) must exist before US-05 (Prevent Double Booking) because the overlap check runs during the search and booking flow. US-04 (Make Reservation) must work before US-06 (Cancel) or US-08 (Modify) because there must be reservations to act on.

Risk and Complexity

US-08 (Modify Reservation) is marked Large because it combines availability checking, date recalculation, and price adjustment in a single transaction with many edge cases. US-05 (Prevent Double Booking) appears small but carries high risk due to concurrency concerns. Both are scheduled early to surface technical challenges before later sprints.

Must-Have vs. Nice-to-Have

US-01 through US-09 form the Minimum Viable Product: guests can search, book, cancel, modify, and view history, while staff can manage daily operations. US-10 through US-12 are enhancements (check-in/out tracking, email notifications, reporting) that improve the experience but are not required for basic reservation management.

Definition of Done

- ☒ Code runs without errors: Feature executes successfully without crashes or unhandled exceptions
- ☒ All acceptance criteria met: Every AC for the story is implemented and verified
- ☒ Manually tested: Feature tested with various inputs including edge cases
- ☒ Code integrated: Feature merged into shared codebase without breaking existing functionality
- ☒ Demonstrated to team: Story shown working during sprint review

Sign-Off

Team Scrum Master: Kyrylo Pryiomyshev _____ Date: March 2, 2026